Interactive Theorem Proving with Isabelle/HOL

Mnacho Echenim, Nicolas Peltier

With contributions from Hervé Guiol, Mehdi Mhalla and several Ensimag students

Grenoble INP - UGA, LIG-CAPP

About this talk

- A quick introduction to interactive theorem proving
- Some transversal formalization efforts
  - **The Cox-Ross-Rubinstein model**
  - The CHSH inequality and Tsirelson's upper-bound
  - The Superposition Calculus
- Current formalization work on AMMs

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- Use automated reasoning tools
  - In the context of Interactive Theorem Proving
  - Explore different formalization tasks

Interactive theorem proving

On formal proofs

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- Why they are important
  - They can be independently verified by any proof checker
  - There can be no gaps or mistakes in a formal proof
  - G. Gonthier: *"all you have to do is agree with the definitions"*

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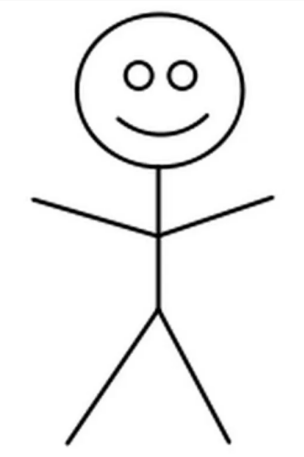
- Why they are important
  - They can be independently verified by any proof checker
  - There can be no gaps or mistakes in a formal proof
  - G. Gonthier: *"all you have to do is agree with the definitions"*
- A formal proof can be very (very, very) long
  - They cannot reasonably be done by hand
  - Theoretical limitations to using automated theorem provers to construct every possible formal proof
  - The trade-off: **interactive theorem provers**

About interactive theorem proving

Principle: collaboration between a human and a program to construct formal proofs



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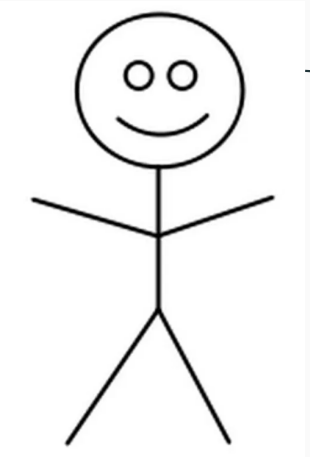

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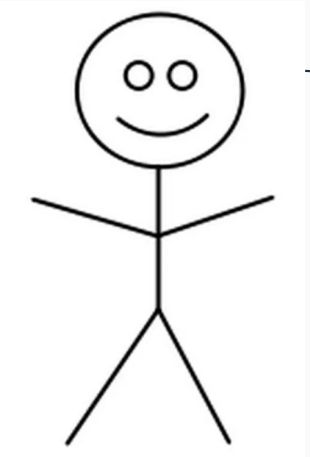

Theorem

Fill proof steps

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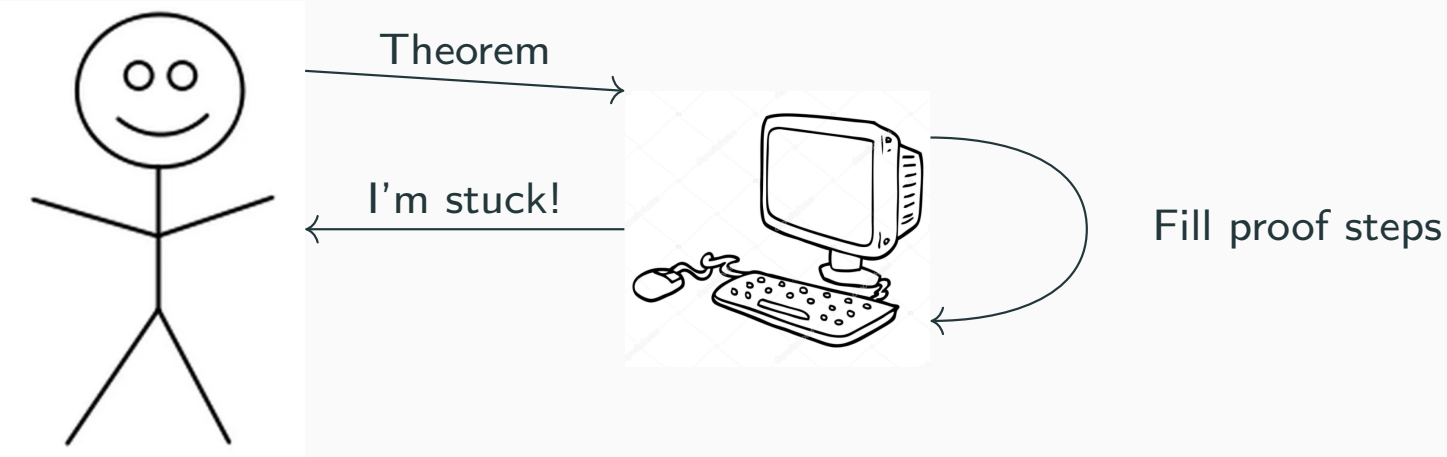

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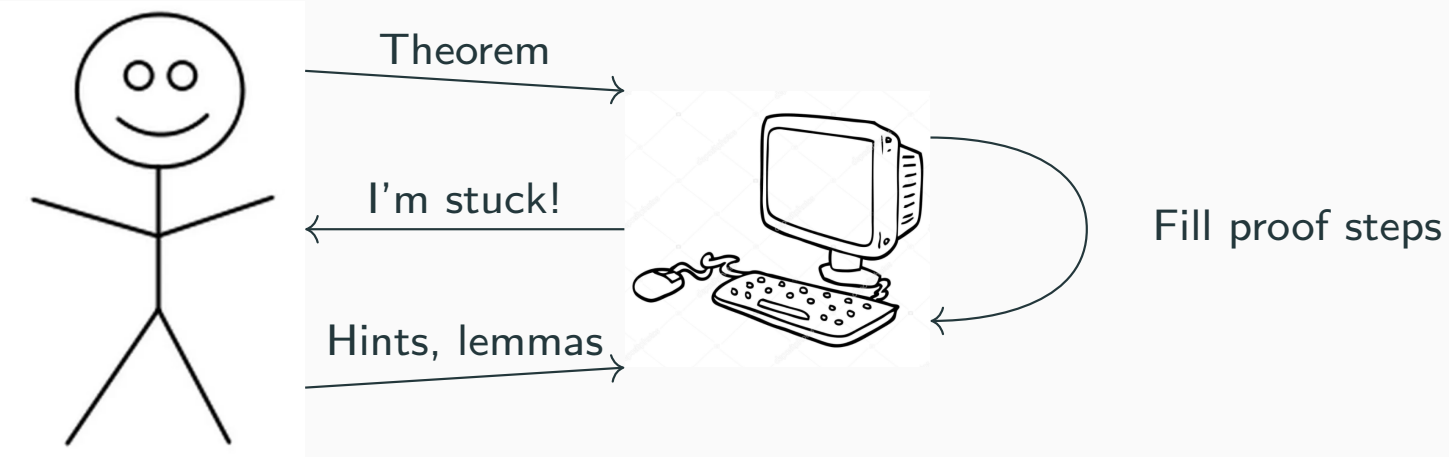

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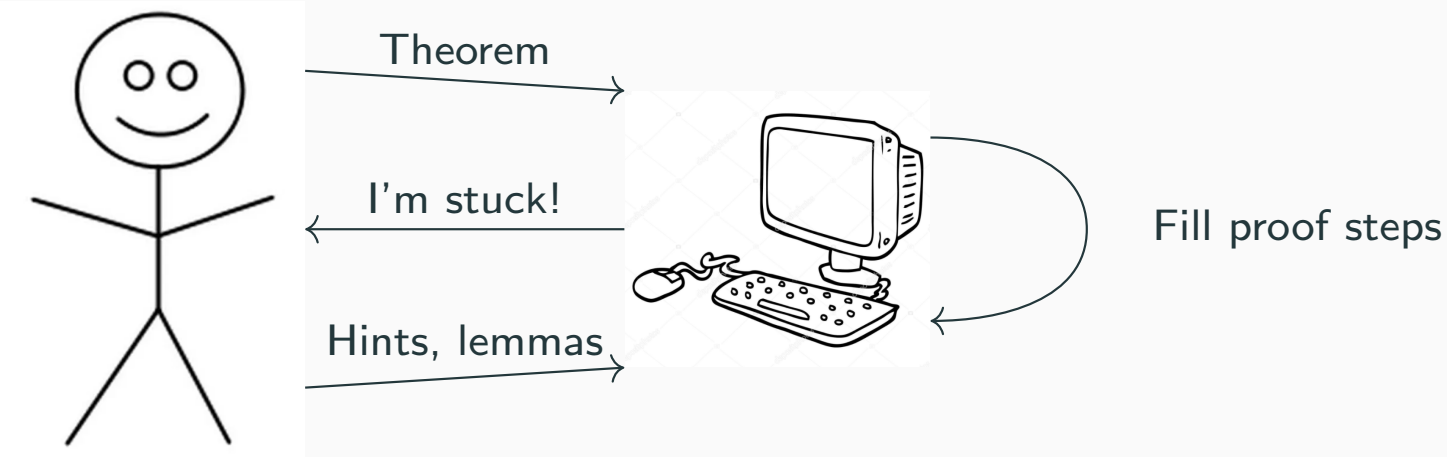

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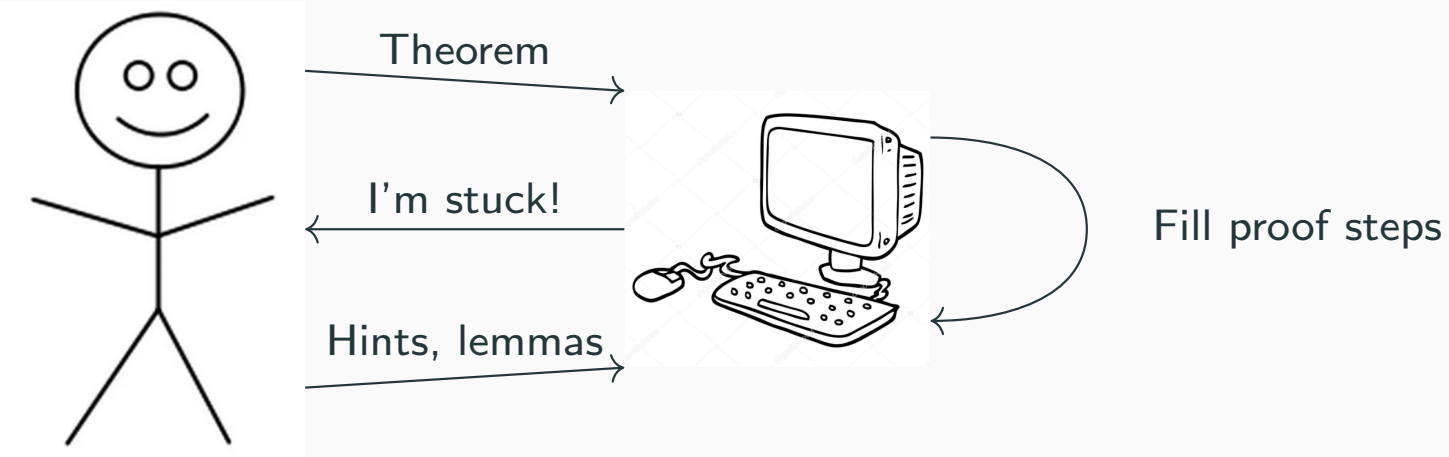

- Other features
  - Theory exploration: *what happens when I change/relax a hypothesis?*
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- Popular systems
  - Coq, **Isabelle**, Lean, ...

Some success stories

- In mathematics
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- In computer science
  - CompCert compiler (used in the Airbus toolchain)
  - sel4 microkernel, used in aviation, spaceflight
  - OpenSSL implementation of Keyed-Hash Message Authentication Code (HMAC)

Correcting mistakes

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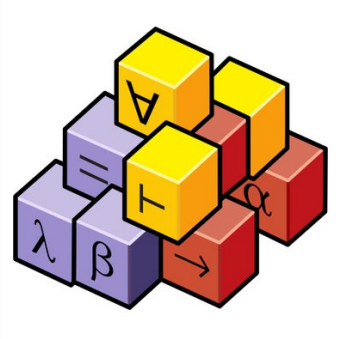

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  - `simp`, `auto`, `blast`, `metis`, `sledgehammer`
- Built-in tools for disproving statements
  - Because many statements we write start out being false (missing hypothesis. . .)
  - `nitpick`, `quickcheck`

The Cox-Ross-Rubinstein model

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    - Arbitrage processes and fair prices
    - Replicating portfolios and complete markets

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  - Derive pricing functions for standard options that are correct by design

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  `market-assets` :: $(\mathbb{N} \to \alpha \to \mathbb{R})$ `set`
- **typedef** α `disc-pf`
  A (discrete) portfolio p contains an evolving quantity of assets. Notations:
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- The value process of a portfolio p containing m assets is

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- A portfolio with m assets is self-financing if $\forall n \geq 1$ and $\forall \omega \in \Omega$:

$$\texttt{val-proc}\ p\ (n+1)\ \omega \ = \ \sum_{i=1}^{m} (\pi_i^p\ (n+1)\ \omega) \cdot (\kappa_i^p\ n\ \omega)$$

Arbitrages, viable markets

$$
\begin{aligned}
&\texttt{arbitrage-process} && :: && (\alpha, \beta)\,\texttt{discrete-market} \to \alpha\ \texttt{disc-pf} \to \mathbb{B} \\
&\texttt{arbitrage-process}\ \texttt{Mkt}\ p && \Leftrightarrow && \exists N \in \mathbb{N}. \\
& && && (\texttt{trading-strat}\ p)\ \wedge\ (\texttt{self-financing}\ p)\ \wedge \\
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&\texttt{viable-market} && :: && (\alpha, \beta)\,\texttt{discrete-market} \to \mathbb{B} \\
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Complete markets and fair prices

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lemma REPLICATING-FAIR-PRICE

assumes `viable-market Mkt`

and `replic-pf` $p\ \kappa\ T$

and `support-adapt Mkt` p

shows `fair-price Mkt (init-value` p`)` $\kappa\ T$

The Cox-Ross-Rubinstein setting

- Probability space: "infinite coin-toss space". Outcomes are sequences $\omega = \langle \omega_1, \ldots, \omega_n, \ldots \rangle$, where every ω_i is $\top$ with probability p and $\bot$ with probability $1 - p$

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  - Risky asset: initial value S_0, $d < u$ and

    $$S\ (n+1)\ \omega\ = \begin{cases} u.(S\ n\ \omega) & \text{if } \omega_n \text{ is } \top \\ d.(S\ n\ \omega) & \text{if } \omega_n \text{ is } \bot \end{cases}$$

Main results

- In the CRR model, a market is viable if and only if $d < 1 + r < u$
- The market is complete in this model

CRR_Model.thy (%ISABELLE_HOME%\afp-2024-01-23\thys\DiscretePricing\)

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definition (in disc_equity_market) complete_market where
"complete_market ⟷ (∀matur. ∀ der∈ borel_measurable (F matur).
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lemma (in CRR_market_viable) CRR_market_complete:
  shows "complete_market" unfolding complete_market_def
proof (intro allI impI)
  fix matur::nat
  show "∀ der ∈ borel_measurable (G matur).
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  proof
    fix der::"bool stream⇒real"
    assume "der ∈ borel_measurable (G matur)"
    define N where "N = bernoulli_stream ((1 + r - d) / (u - d))"
    hence "replicating_portfolio (delta_hedging N der matur) der matur"
      using delta_hedging_replicating
      ‹der ∈ borel_measurable (G matur)› by simp
    thus "∃pf. replicating_portfolio pf der matur" by auto
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- The entire formalization is available on the *Archive of Formal Proofs*
  - `https://www.isa-afp.org/entries/DiscretePricing.html`
  - The entire formalization is over 350 pages long, even using Isabelle's automation

Other formalization efforts

On local hidden variables

Is quantum mechanics complete?

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  - Disturbed by statistical nature of measurement outcomes
  - EPR paradox: *spooky action at a distance*

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- The existence of such models is refuted by Bell's inequality (1964)
  - CHSH inequality: **Proposed experiment to test local hidden-variable theories** [Clauser, Horne, Shimony, Holt, 1969]
  - One of Bell's inequalities that was verified experimentally by A. Aspect

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  - One of Bell's inequalities that was verified experimentally by A. Aspect
- Formalization of this inequality and the accompanying Tsirelson bound
  - `https://www.isa-afp.org/entries/TsirelsonBound.html`
  - Total formalization (along with fundamental quantum notions) is over 250 pages long

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- N. Peltier: formalization of the Superposition Calculus *within* Isabelle
    - `https://www.isa-afp.org/entries/SuperCalc.html`

Ongoing work on AMMs

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Thank you for your attention